Red Torque – Detailed Major Component Study

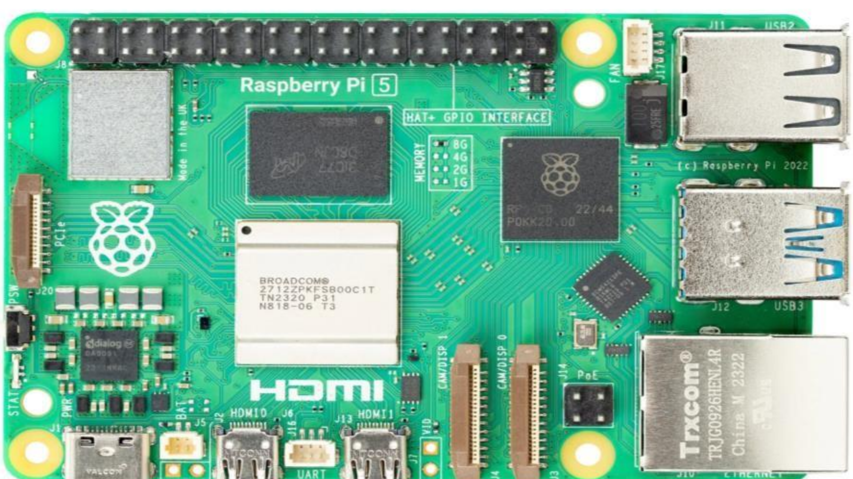
Date: 2 May 2026

1. Raspberry Pi 5

The Raspberry Pi 5 is the latest and most powerful version of the Raspberry Pi computer series. It looks like a small board but has the power of a full desktop. It is made for DIY and educational projects, especially robotics.

In our robot project, it will be the main brain handling image processing, decisionmaking, and controlling other components.

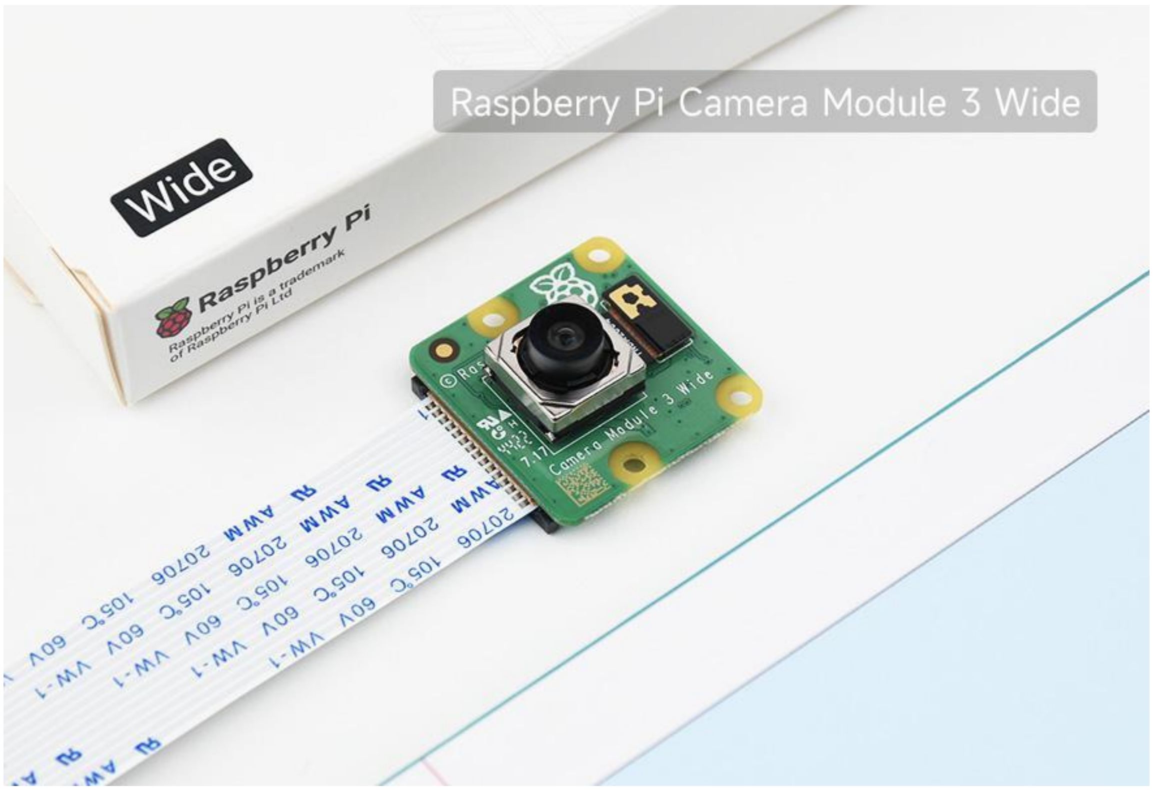
- Quad-core 64-bit ARM Cortex-A76 CPU running at 2.4GHz – faster than Pi 4.
- Dual camera ports – supports two camera modules at the same time.
- PCIe 2.0 interface – allows fast SSD storage or expansion cards.
- Real-time clock – keeps track of time even without power.
- Improved GPIO performance – faster and better I/O for sensors.
- Supports active cooling via fan connector directly on board.
- USB 3.0 ports – faster data transfer from external devices.
- Used as the main controller to run Linux (Raspberry Pi OS).



The Raspberry Pi 5 acts as the main brain of the robot, handling tasks like image processing, sensor integration, and decision-making. It runs the software that controls movement, processes camera input, and communicates with other modules like the ESP32.

2. Microsoft USB webcam

The **Raspberry Pi Camera Module 3 Wide** is an advanced camera designed specifically for Raspberry Pi boards, offering improved image quality and a wider field of view for embedded vision applications.



• Provides up to **12MP resolution** using the Sony IMX708 sensor, supporting high-quality image capture and video recording.
• Offers a **120° wide field of view**, making it ideal for applications requiring broader scene coverage such as robotics navigation and surveillance.
• Supports **HDR (High Dynamic Range)** for improved performance in varying lighting conditions.
• Includes **autofocus capability**, enabling sharp image capture without manual lens adjustment.
• Connects via the **CSI (Camera Serial Interface)** port, ensuring high-speed, low-latency data transfer compared to USB webcams.
• Fully compatible with **OpenCV**, **libcamera**, and Python-based image processing

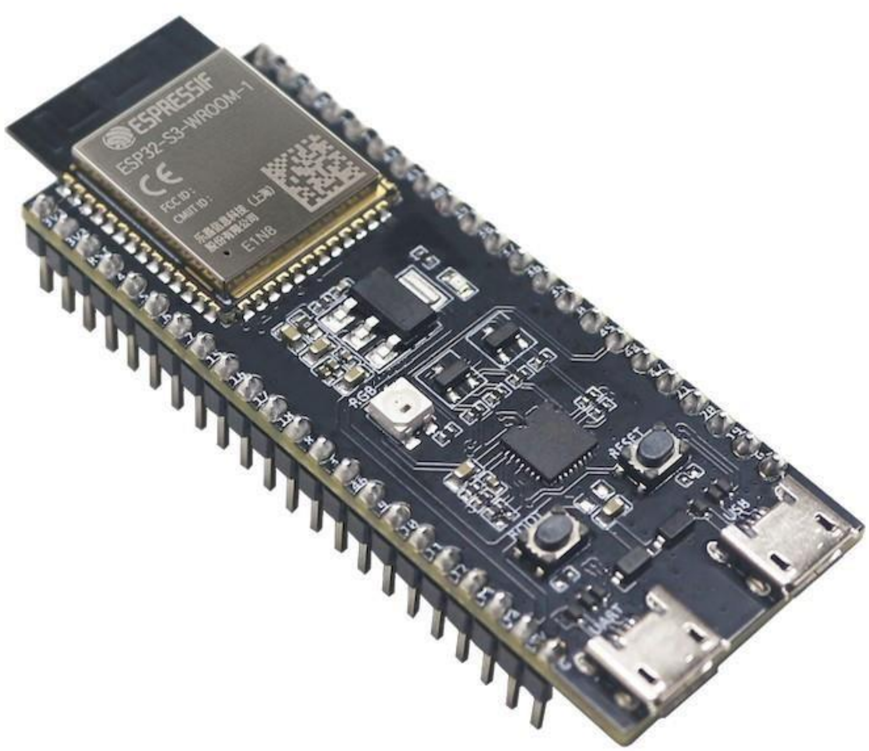
workflows.
• Suitable for real-time applications such as object detection, lane/line following, and spatial awareness in robotic systems.

3. ESP32 Microcontroller

The ESP32 is a low-cost, low-power microcontroller board that includes built-in Wi-Fi and Bluetooth. It is popular in IoT and robotics projects.

- Dual-core processor with speeds up to 240 MHz.
- Wi-Fi and Bluetooth capabilities make it suitable for wireless communication.
- Can be programmed using Arduino IDE or MicroPython.
- Multiple GPIO pins to connect sensors and control motors.
- I2C, SPI, UART interfaces for communication with other boards and modules.
- In our project, used as a helper board to offload simple sensor readings or Bluetooth tasks.

The ESP32 is used for handling low-level tasks like motor control, reading sensor data, and wireless communication (Bluetooth or Wi-Fi). It works alongside the Raspberry Pi to offload real-time processing and ensures smooth coordination between hardware components in the robot.



5. Buck Converter & USB Buck

A buck converter is a voltage step-down converter that changes a high input voltage to a lower output voltage. It is crucial for safely powering electronics from batteries.

- Converts 11.1V from battery to 5V or 3.3V for Raspberry Pi and ESP32. • Some models come with USB output for easy powering of USB devices.
- Adjustable voltage using a screw trimmer.
- Ensures consistent and safe voltage levels to prevent board damage.
- Two types used: standard screw terminal buck and USB buck module.

The buck converter steps down the high voltage from the battery (like 12V) to a lower voltage (like 5V) needed by the Raspberry Pi and other components. The USB buck converter does the same but gives output through a USB port, making it easier to power boards like the Pi or ESP32 directly.

USB BUCK CONVERTER **DC-DC BUCK CONVERTER**

5. Battery

The **GenX Power Premium 7.4V LiPo Battery** is a high-performance rechargeable battery commonly used in robotics, drones, and embedded systems requiring stable and high current output.

• Available in multiple capacities such as 650mAh, 1000mAh, 2200mAh, up to 8000mAh, allowing flexibility based on power requirements.
• Operates at a **nominal voltage of 7.4V (2S configuration)**, meaning two cells connected in series for stable output.
• Provides high **discharge rates (25C–80C)**, enabling it to supply large currents required for motors, drivers, and high-load components.
• Lightweight and compact design with **high energy density**, making it ideal for mobile robots and UAV systems.
• Typically comes with connectors like **XT60**, ensuring reliable and low-resistance power delivery.
• Built using matched cells for **consistent performance and longer usable runtime per charge cycle**.
• Suitable for applications such as robotics, RC vehicles, drones, and embedded projects requiring portable power.

6. Lego SPIKE Prime set (Chassis and inbuilt microcontroller)

The **LEGO SPIKE Prime (Chassis & Inbuilt Microcontroller)** is an integrated robotics platform that combines a modular mechanical structure with a programmable control hub, enabling rapid prototyping and autonomous robot development.

• Consists of over **500+ LEGO Technic elements**, enabling construction of a customizable and reconfigurable robot chassis.
• Includes a **programmable Intelligent Hub (microcontroller)** based on STM32 architecture, acting as the brain of the system.
• The hub features **6 input/output ports**, allowing connection of motors and sensors for full robotic control.
• Built-in components include a **5×5 LED matrix, Bluetooth connectivity, speaker, and 6-axis gyro sensor** for feedback and interaction.
• Supports both **block-based (Scratch) and Python programming**, enabling scalable learning from beginner to advanced robotics.
• The chassis uses **LEGO Technic beams, gears, wheels, and connectors**, providing strong mechanical structure for mobility and manipulation tasks.
• Comes with **motors (1 large + 2 medium)** and sensors like **color, distance, and force**, enabling real-world interaction and automation.
• Designed for **modular robot design**, allowing quick iteration of wheeled robots, arms, and autonomous systems.

Summary

The core components explored include the Raspberry Pi 5, which functions as the main processor responsible for handling computation and data processing. The Microsoft USB Webcam is used for capturing images and video, making it suitable for vision-based tasks. The ESP32 microcontroller provides wireless connectivity through Wi-Fi and Bluetooth and supports multiple I/O functions for flexible hardware interfacing. A 2200mAh LiPo battery serves as the main power source, supplying energy to all components. Voltage regulation is managed through a buck converter and a USB buck converter, which safely step down higher voltage to levels required by sensitive electronics.

Each component plays a vital role in building a stable and efficient electronic system. Their individual functions and purposes are clearly identified and will be used collectively in the system's design and integration.